MIL-T-51157A
21 April 1967
SUPERSEDING
MIL-T-51157(MU)
25 October 1963

MILITARY SPECIFICATION

TANK ASSEMBLY, FUEL AND AIR; FOR M10-8 FLAME THROWER

This specification is mandatory for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers fuel and air tank assembly for use in the M10-8 flame thrower.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

DL81-1-9063 - Tank, Fuel and Air.

(All drawings referenced in this specification are listed on above Edgewood Arsenal Dwg DL81-1-9063.)

FSC 1040

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL81-1-9063 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the fuel and air tank shall conform to the specifications and drawings listed on Drawing DL81-1-9063 and subsidiary drawings.

3.2 Assembly. The fuel and air tank shall be assembled as shown on Drawing E81-1-9063.

3.3 Permanent volumetric expansion. The permanent volumetric expansion of the fuel tank (Dwg E81-1-7467) and the air tank (Dwg D81-1-7468) shall not exceed 10 percent of the total volumetric expansion when tested as specified in 4.2.2.1.

3.4 Leakage. The fuel tank shall not exceed 15 pounds per square inch gage (psig) and the air tank 100 psig loss of air pressure when tested as specified in 4.2.2.2.

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3.5 Workmanship. The fuel and air tank shall be free from damage, which could affect subsequent assembly, and foreign matter such as metal particles, filings, grease or corrosion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratories acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Inspection provisions.

4.2.1.1 Examination and nondestructive tests. Each fuel tank and air tank shall be inspected in accordance with the list of defects. If any tank contains any defect listed in 4.2.1.2, that tank shall be rejected.

4.2.1.2 List of defects.

(a) Tank, air (Dwg D81-1-7468).

1	Outside diameter incorrect
*2	Tee threads incorrect
3	Volumetric expansion (4.2.2.1)
4	Leakage (4.2.2.2)
5	Protective finish missing or incorrect
6	Workmanship (3.5)
7	Serial number missing
8	Component missing or incorrectly assembled

*For repair parts only.

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(b) Tank, fuel (Dwg E81-1-7467).

1 Component missing or incorrectly assembled
2 Tube location incorrect
3 Nozzle outside diameter incorrect
4 Nozzle undercut location incorrect
5 Nozzle location incorrect
6 Foot plate, hole location incorrect
7 Foot plate location incorrect
8 Non-parallelism excessive
9 Machine finish incorrect
10 Workmanship (3.5)
11 Volumetric expansion (4.2.2.1)
12 Leakage (4.2.2.2)
13 Protective finish missing or incorrect
14 Serial number missing

4.2.2 Tests.

4.2.2.1 Permanent volumetric expansion. The permanent volumetric expansion of the fuel and air tanks shall be determined by water jackets or other suitable methods to obtain accurate data. The air tank, when subjected to an internal water pressure of 5000 ± 50 psig, shall conform to 3.3 (see 6.2 and 6.3). The fuel tank, when subjected to an internal water pressure of 550 ± 20 psig, shall conform to 3.3. The pressure shall be held for a minimum period of 1 minute after it is stabilized. The pressure gage shall be accurate to permit reading of 1 percent and the expansion gage to an accuracy of either 1 percent or 0.1 cubic centimeter. Prior to this test the air tank shall not be subjected to a pressure greater than 4500 psig and the fuel tank shall not be subject to a pressure greater than 495 psig.

4.2.2.2 Leakage. The fuel tank, when subjected to an internal air pressure of 325 ± 5 psig, shall conform to 3.4. The air tank, when subjected to an internal air pressure of 3000 ± 50 psig, shall conform to 3.4 (see 6.2). The pressure shall be held for a minimum period of 5 hours. The pressure in the fuel and air tanks shall be corrected to standard conditions of temperature and air.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The fuel and air tank shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

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5.2 Packing, level C. The fuel and air tanks, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When packaging data sheet is required.

6.2 Caution. Due to the hazard of testing with high pressure, suppliers should provide adequate protection for personnel while testing in accordance with this specification.

6.3 Test apparatus failure. If the test pressure cannot be maintained for a sufficient period of time to determine permanent volumetric expansion, due to failure of the test apparatus, the test may be repeated at a pressure increased by 100 psig when testing the air tank and 20 psig when testing the fuel tank.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)
Project No. 1040-0084

User activity:

Air Force - 84

Coordination Method Code: N

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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